MATHEMATICS PROJECT WORK

Formative Assessment 1 (FA 1)

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1. Name of the Project

Finding H.C.F. by using Euclid's Division Lemma.

2. Collection of Data: History of Euclid

- **Father of Geometry** (approx. 300 B.C.)
- **Field:** Geometry
- **Birth place:** Alexandria of Egypt

Noted theories:

- He introduced the first system of thought based on fundamental definitions, propositions, and rules of inference or logic.
- Euclid invented a new way of finding H.C.F. through *Euclid's Division Lemma*.
- He wrote 13 books collectively called **"THE ELEMENTS"**.

3. Objectives

To find the H.C.F. of the two numbers i.e., **8** and **14** experimentally based on Euclid's division lemma.

4. Materials Required

- Cardboard sheet
- Glazed coloured papers (or Graph sheet)
- Sketch pens
- Scissors
- Geometry Box
- Fevicol

5. Pre-requisite Knowledge

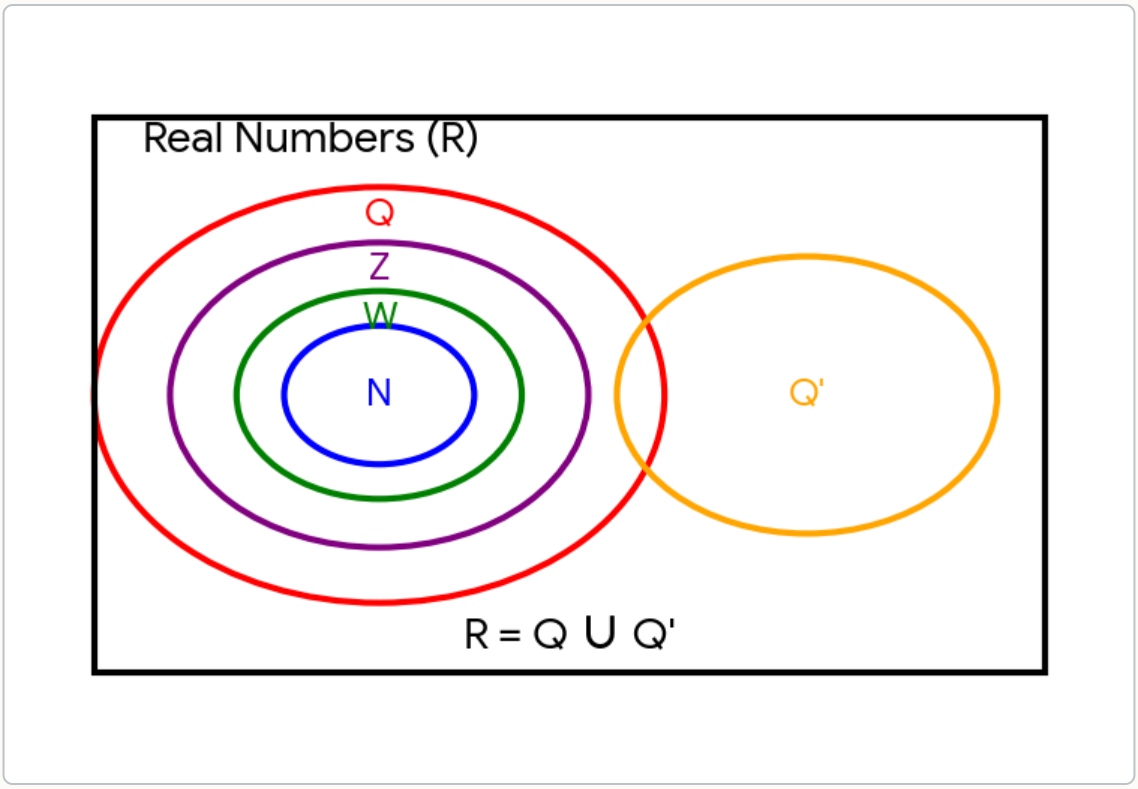


Fig 1: Real Numbers Classification

- **Real Numbers:** $R = Q \cup Q'$
- **Lemma:** A lemma is a proven statement used for proving another statement.
- **Algorithm:** An algorithm is a series of well-defined steps which give a procedure for solving a type of problem.
- **Euclid's Division Lemma:** Let a and b be any two positive integers. Then there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < b$.
- **Euclid's Division Algorithm:** If a and b are two positive integers such that $a = bq + r$, then every common divisor of a and b is a common divisor of b and r.
- **H.C.F:** Highest Common Factor.

6. Procedure

- Take a cardboard sheet of a convenient size.
- Cut one strip of length a units, cut one strip of length b units ($b < a$).
- Cut two strips each of length c units ($c < b$).
- Cut one strip of length d units ($d < c$) and two strips of length e units from the cardboard ($e < d$).

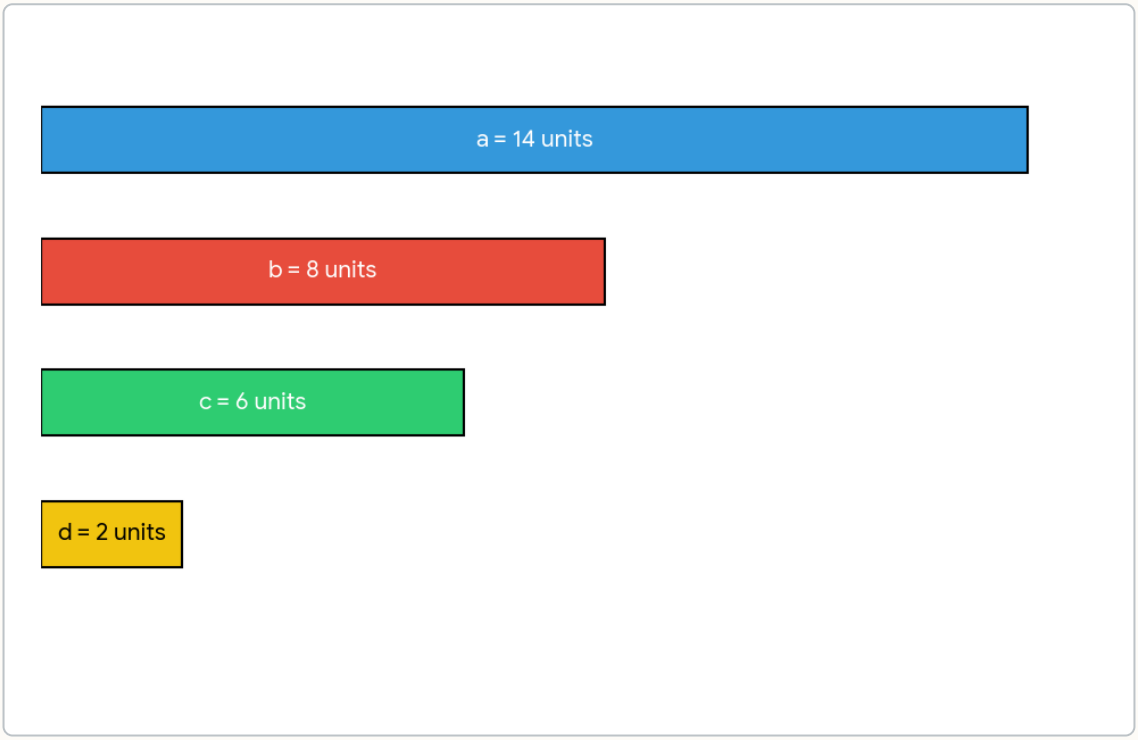


Fig 2: Cardboard strips of various lengths

- Paste glazed papers of different colours on the strips.
- Stick the strips on the paper to find the H.C.F of 14 and 8.

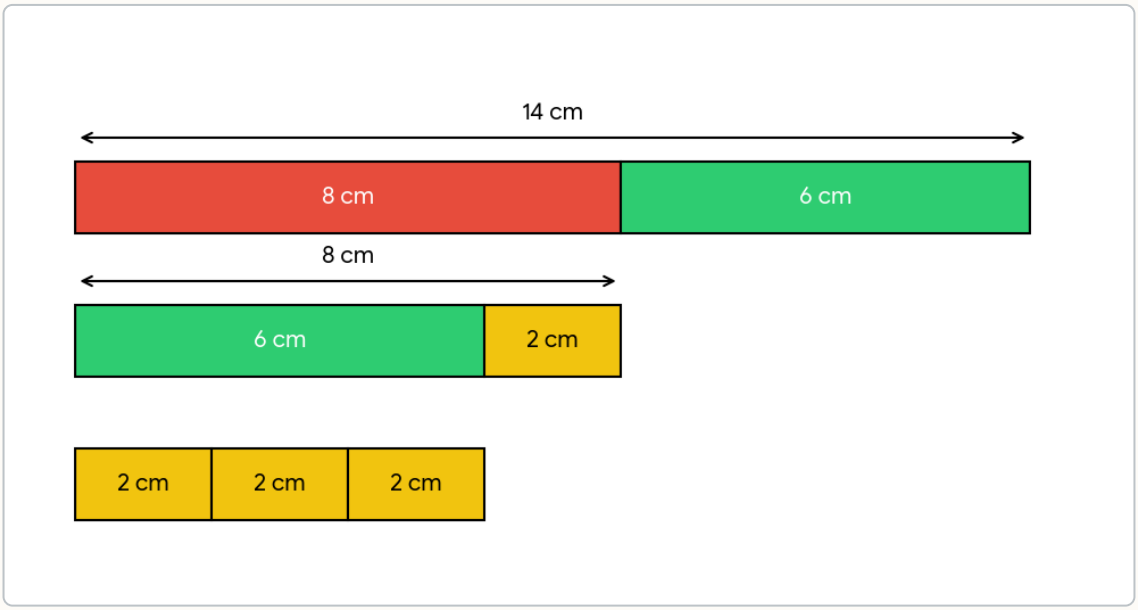


Fig 3: Arranging strips to demonstrate Euclid's Division Lemma

7. Observation

- When comparing the 14 cm strip (a) with the 8 cm strip (b), we observe that one 8 cm strip fits into the 14 cm strip, leaving a gap of 6 cm.

  $$14 = 8 \times 1 + 6$$

- Next, comparing the 8 cm strip with the 6 cm gap (c), we see that one 6 cm strip fits into the 8 cm strip, leaving a gap of 2 cm.

  $$8 = 6 \times 1 + 2$$

- Finally, comparing the 6 cm strip with the 2 cm gap (d), we find that exactly three 2 cm strips fit into the 6 cm strip with no gap left.

  $$6 = 2 \times 3 + 0$$

- Since the remainder is 0 when the strip of length 2 cm is used, the process stops here.

8. Conclusion

The last divisor which leaves no remainder is **2**. Therefore, the Highest Common Factor (H.C.F.) of 14 and 8 is experimentally found to be **2**. This project practically verifies Euclid's Division Lemma.